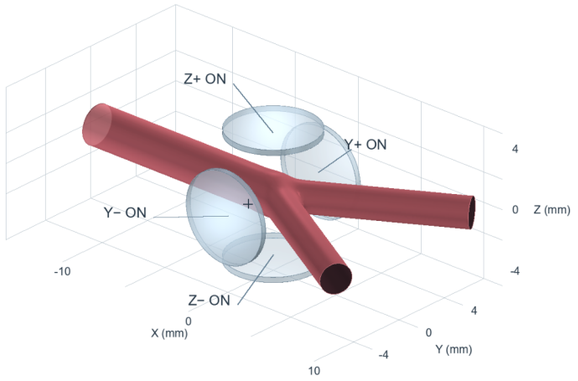
Bubble–bubble attraction enables aggregation

Matched initial positions and PZT drive; only Secondary Bjerknes force is switched OFF

PZT layout and vessel context

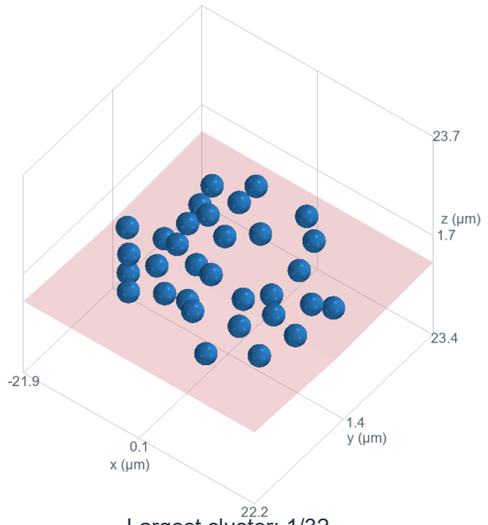


Four discs: 6 mm diameter
Centres: Y = ±4 mm, Z = ±4 mm
Cross: observation location

Local panels share a fixed camera and physical scale.

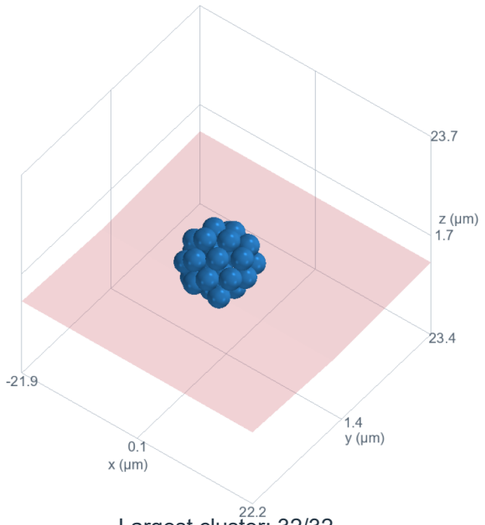
All four PZT elements stay ON.

Same dispersed initial state



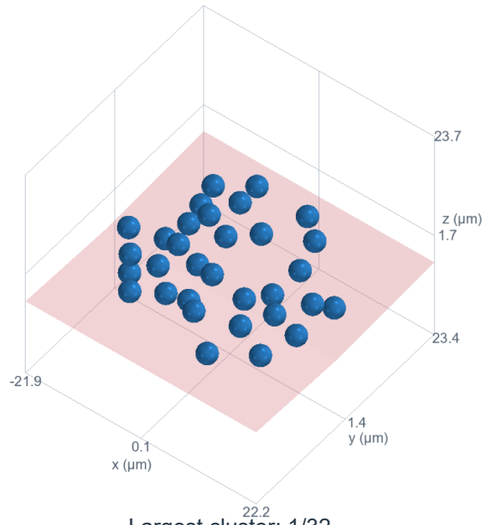
Largest cluster: 1/32

Attraction ON · 600 ms

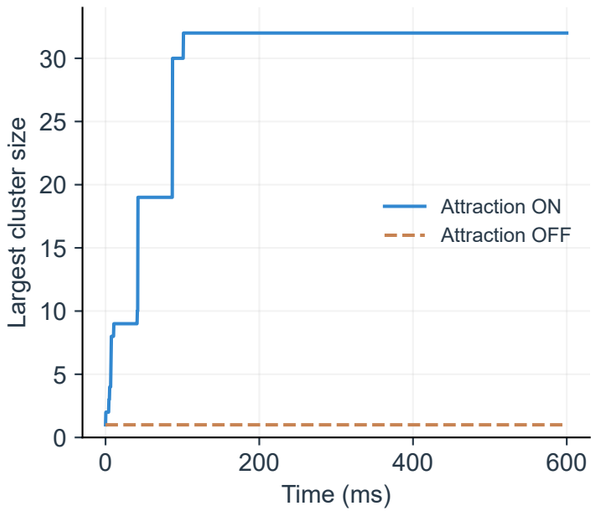


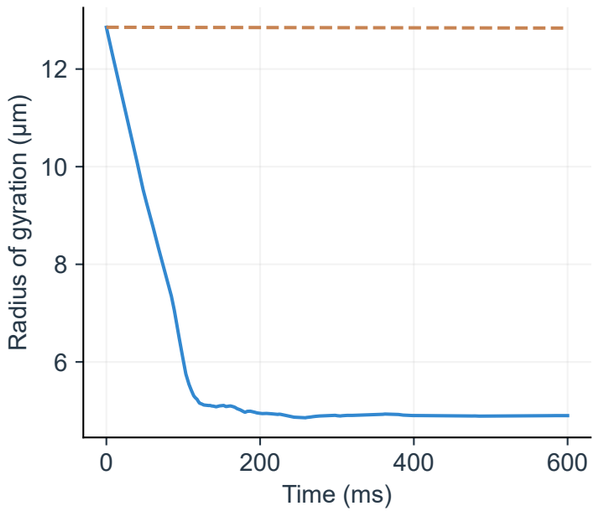
Largest cluster: 32/32

Attraction OFF · 600 ms



Largest cluster: 1/32





ON: 32-bubble cluster

OFF: 32 separate bubbles

PZT remains ON in both cases.
Observation window: 600 ms.
No flow; geometric WALL only.

All 32 actual bubbles • R = 1.78 μm • Local views retain physical relative sizes • EXPLORATORY